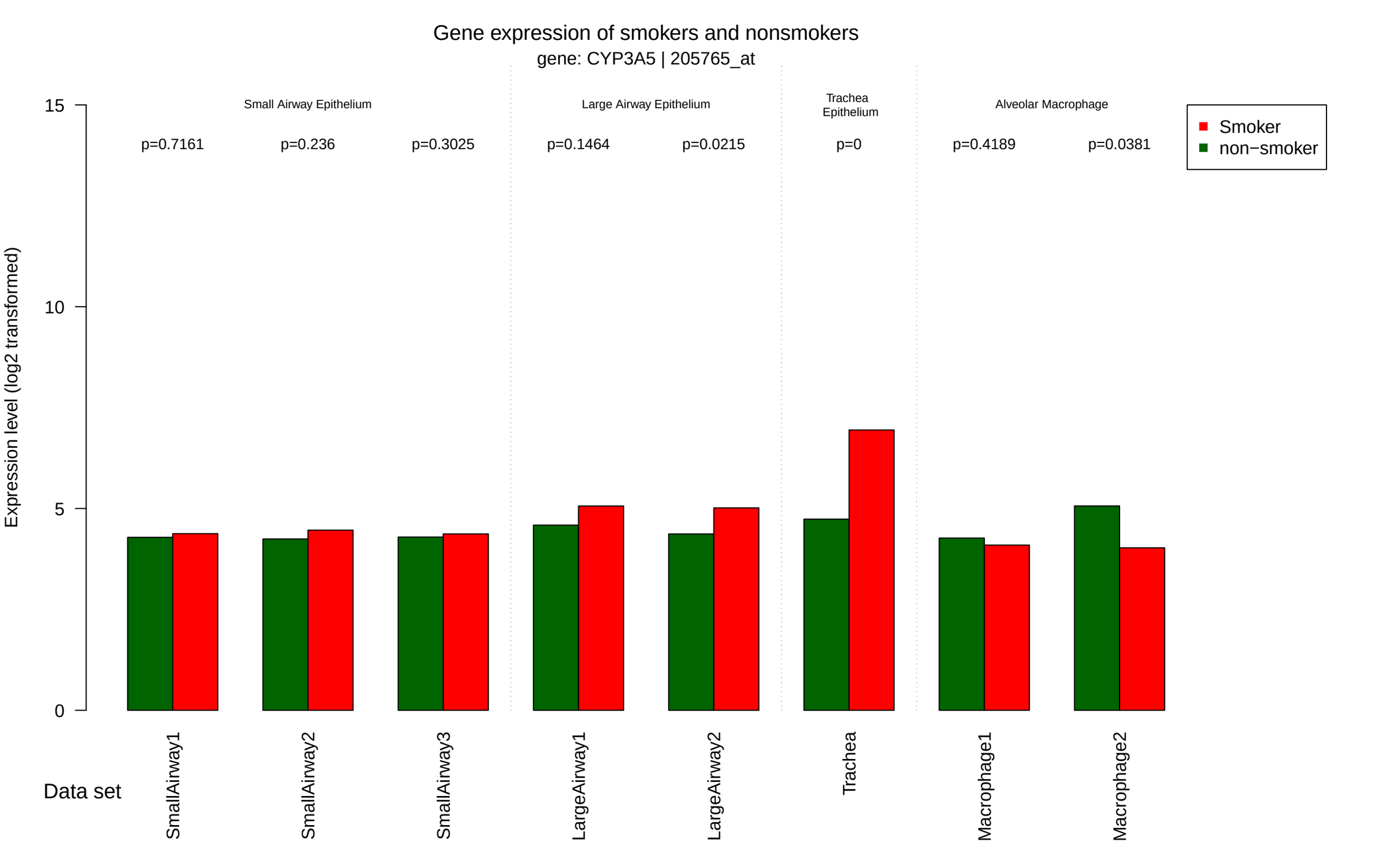
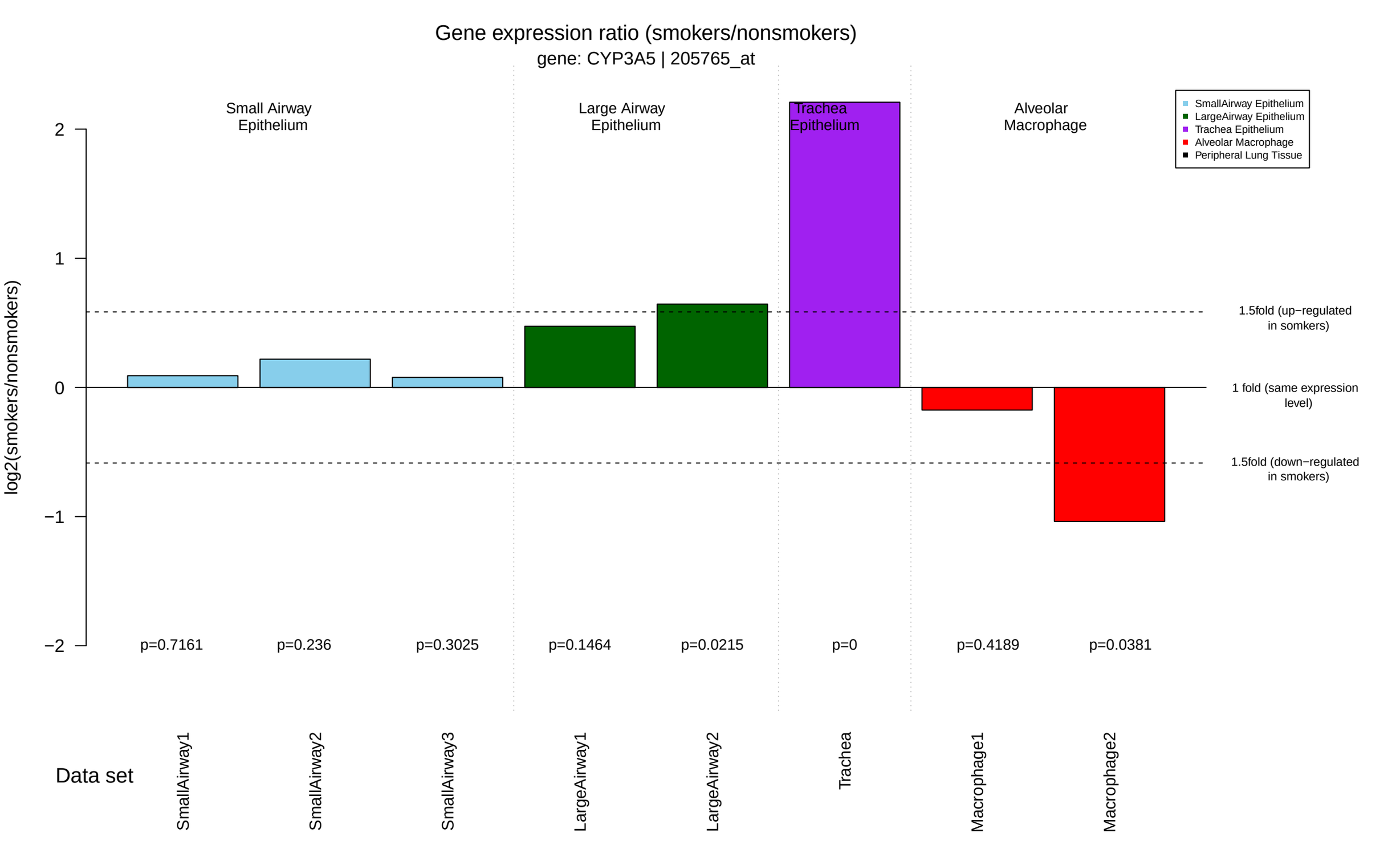
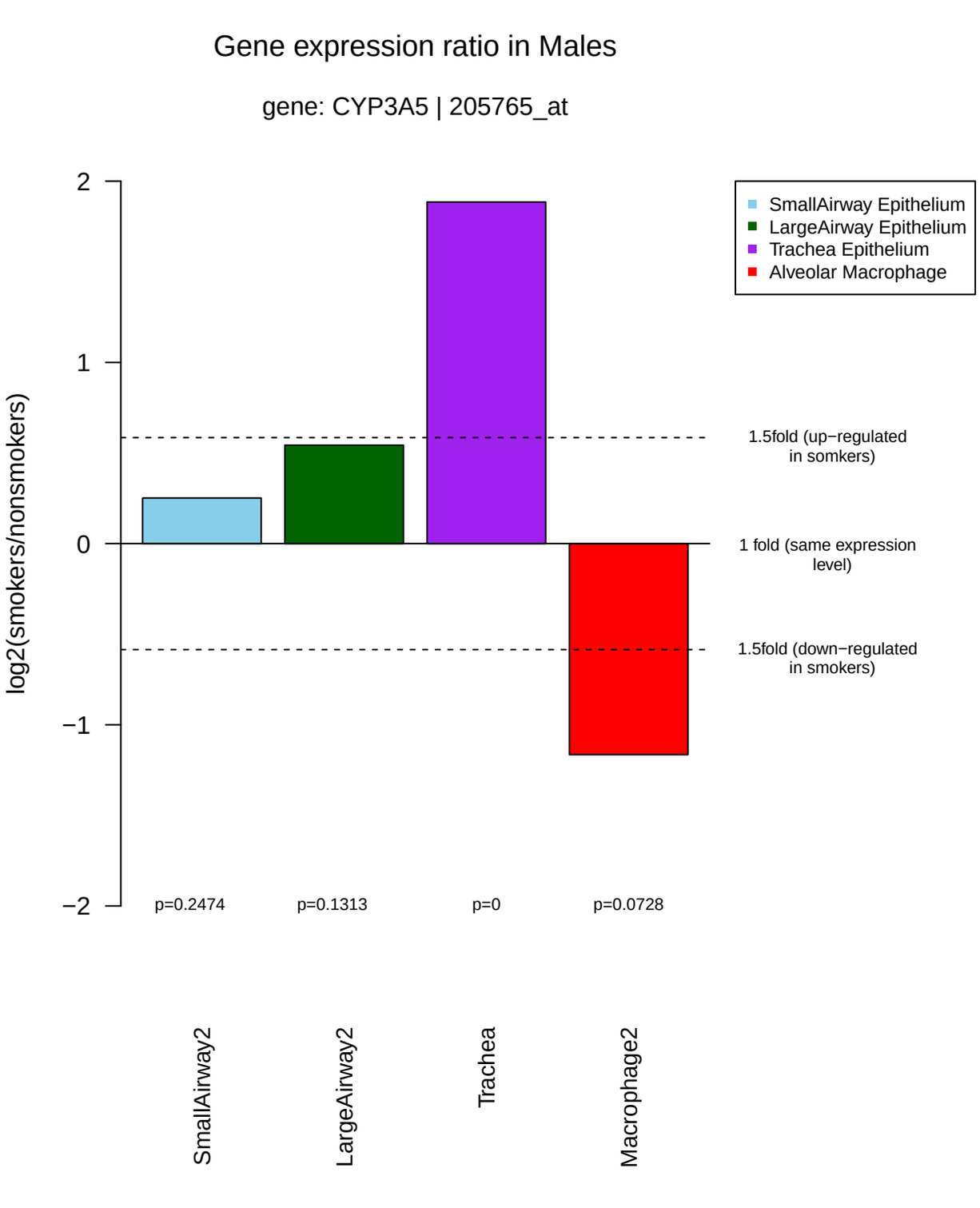
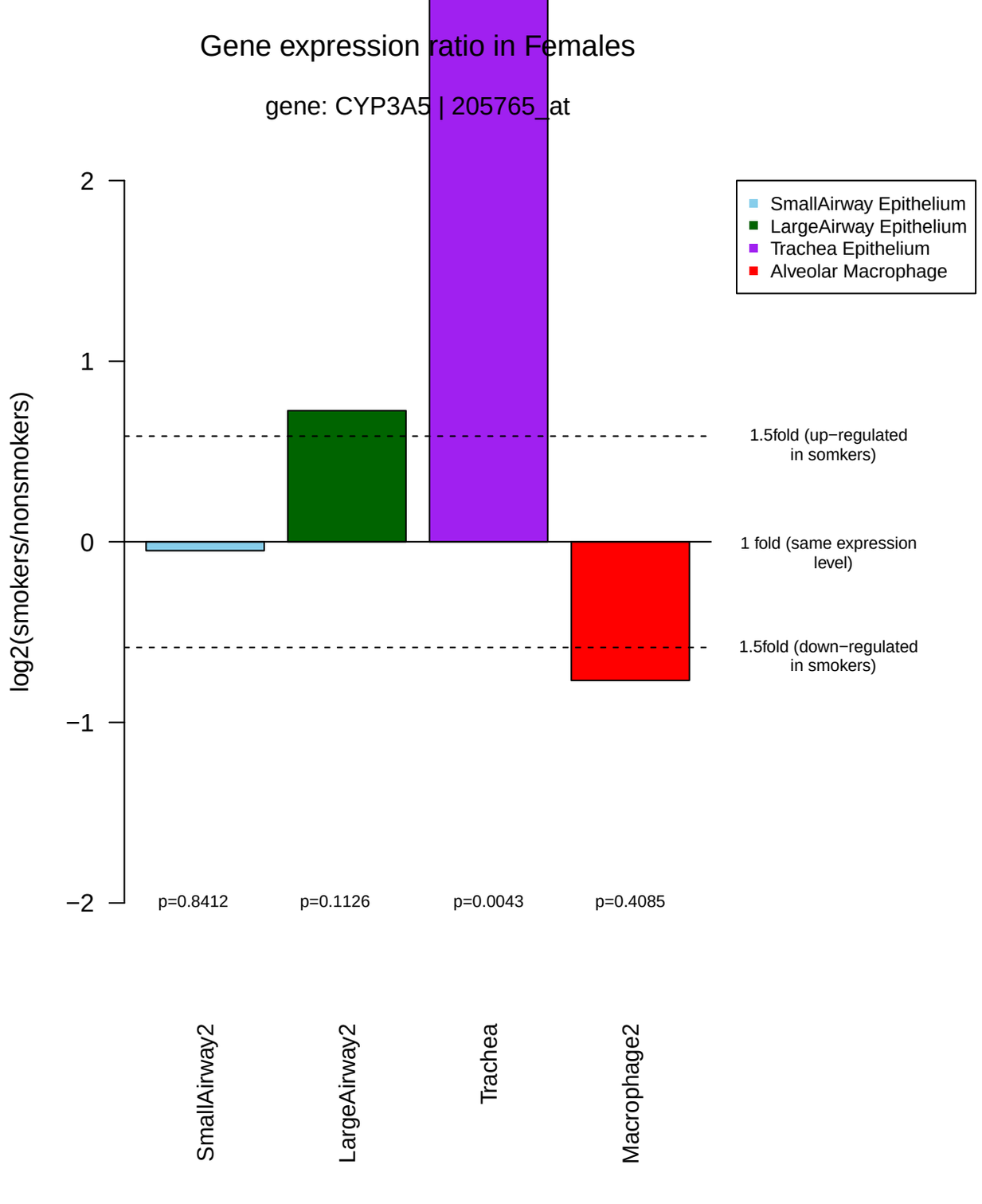
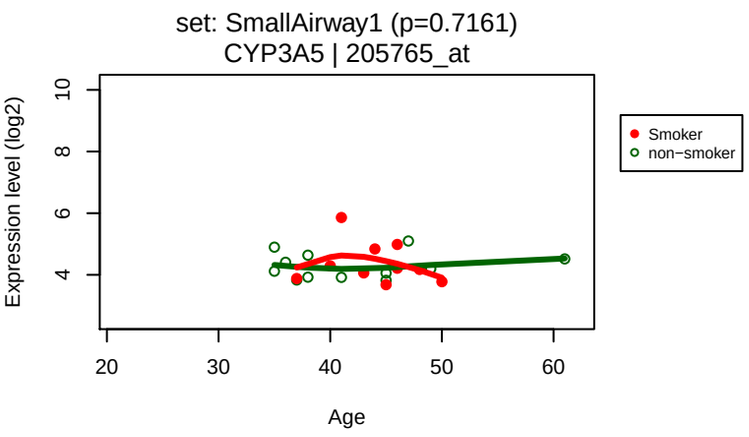
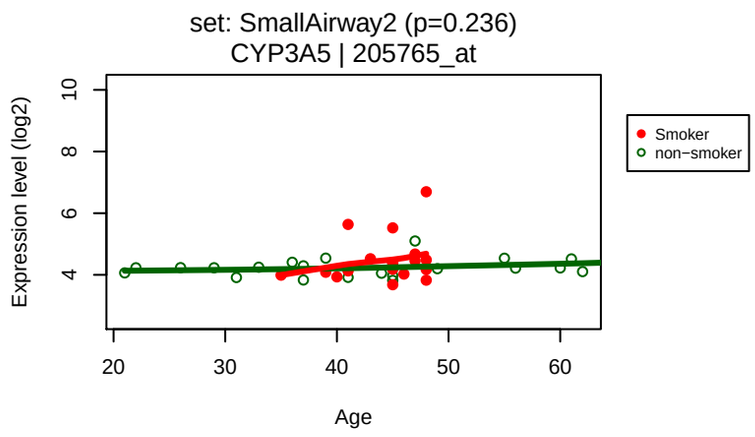

Gene expression of smokers and nonsmokers
gene: CYP3A5 | 205765_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7161
p=0.236
p=0.3025
p=0.1464
p=0.0215
p=0
p=0.4189
p=0.0381
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

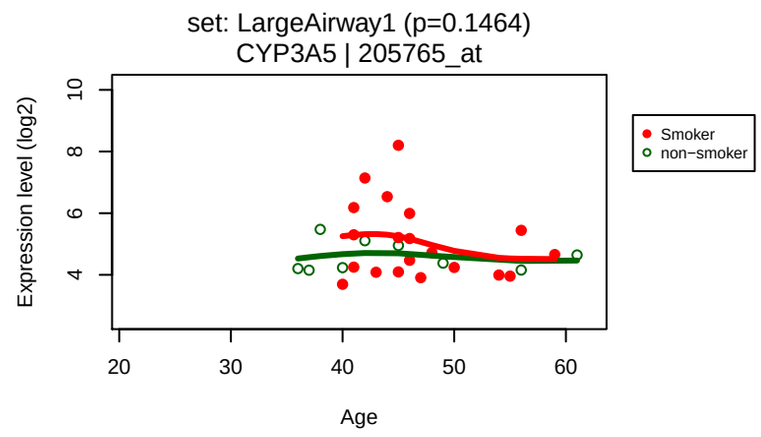

Gene expression ratio (smokers/nonsmokers)
gene: CYP3A5 | 205765_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7161
p=0.236
p=0.3025
p=0.1464
p=0.0215
p=0
p=0.4189
p=0.0381
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

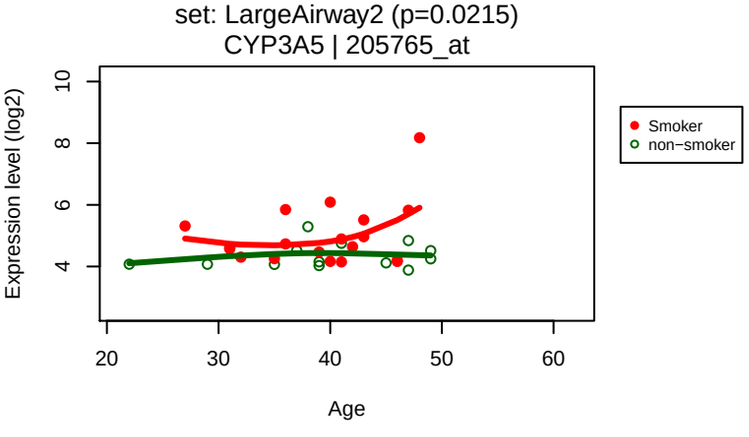

Gene expression ratio in Males
gene: CYP3A5 | 205765_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2474
p=0.1313
p=0
p=0.0728
SmallAirway2
LargeAirway2
Trachea
Macrophage2

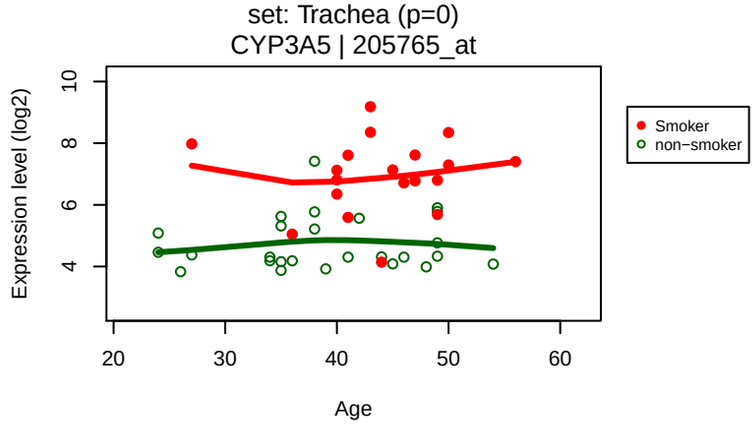

Gene expression ratio in Females
gene: CYP3A5 | 205765_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8412
p=0.1126
p=0.0043
p=0.4085
SmallAirway2
LargeAirway2
Trachea
Macrophage2

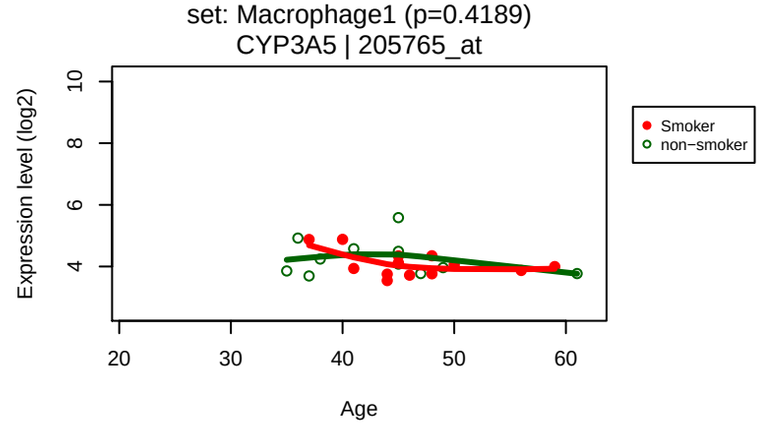

set: SmallAirway1 (p=0.7161)
CYP3A5 | 205765_at
Expression level (log2)
Age
Smoker
non−smoker

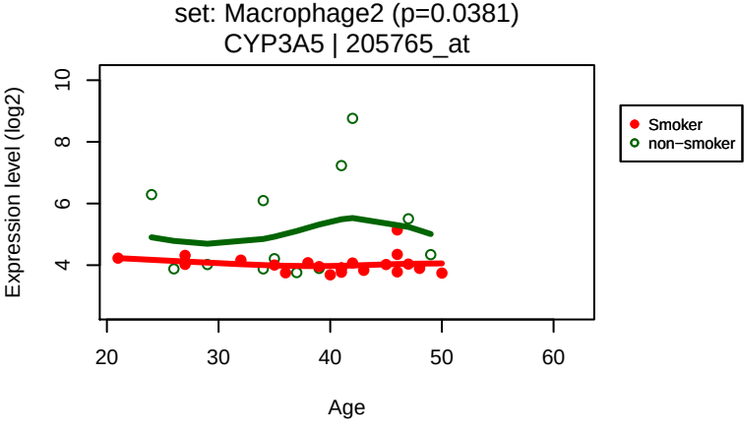

set: SmallAirway2 (p=0.236)
CYP3A5 | 205765_at
Expression level (log2)
Age
Smoker
non−smoker

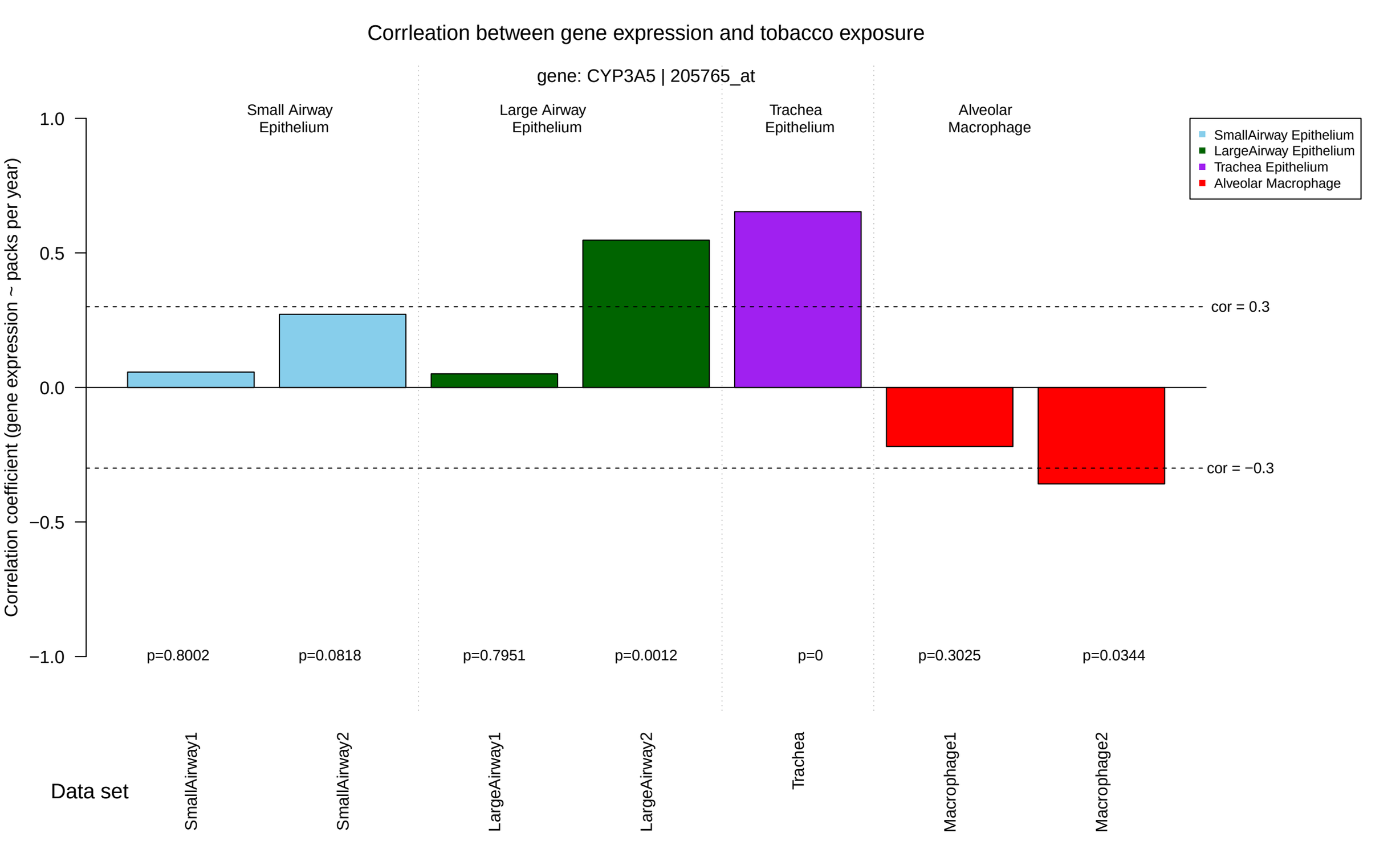

set: LargeAirway1 (p=0.1464)
CYP3A5 | 205765_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0215)
CYP3A5 | 205765_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
CYP3A5 | 205765_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4189)
CYP3A5 | 205765_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0381)
CYP3A5 | 205765_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP3A5 | 205765_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8002
p=0.0818
p=0.7951
p=0.0012
p=0
p=0.3025
p=0.0344
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2